

Putting cancer to sleep at night: the neuroendocrine/circadian melatonin signal.

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Physiological and pharmacological blood concentrations of melatonin inhibit tumorigenesis in a variety of in vivo and in vitro experimental models of neoplasia. Evidence indicates that melatonin's anticancer effects are exerted via inhibition of cell proliferation and a stimulation of differentiation and apoptosis. A new mechanism by which physiological and pharmacological blood levels of melatonin inhibit cancer growth in vivo via a melatonin-induced suppression of tumor linoleic acid (LA) uptake and its metabolism to the important mitogenic signaling molecule 13-hydroxyoctadecadienoic acid (13-HODE). Melatonin suppresses cAMP formation and inhibits tumor uptake of LA and its metabolism to 13-HODE via a melatonin receptor-mediated mechanism in both tissue-isolated rat hepatoma 7288 CTC and human breast cancer xenografts. It has been postulated that in industrialized societies, light at night, by suppressing melatonin production, poses a new risk for the development of breast cancer and, perhaps, other cancers as well. In support of this hypothesis, light during darkness suppresses nocturnal melatonin production and stimulates the LA metabolism and growth of rat hepatoma and human breast cancer xenografts. Nocturnal dietary supplementation with melatonin, at levels contained in a melatonin-rich diet, inhibits rat hepatoma growth via the mechanisms described above. The nocturnal melatonin signal organizes tumor metabolism and growth within circadian time structure that can be further reinforced by appropriately timed melatonin supplementation. Dietary melatonin supplementation working in concert with the endogenous melatonin signal has the potential to be a new preventive/therapeutic strategy to optimize the host/cancer balance in favor of host survival and quality of life.